AGENTIC AI & FULL-STACK APPLICATION DEVELOPMENT

A two-day hands-on workshop
at SRM University

The core promise of this workshop. By the end of today, you will have built and deployed a real web app with an AI agent inside it.

Meet your guide for the next two days

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Focus Areas:

- AI-Assisted Software Development
- Agentic AI Integration
- Enterprise Learning Programs

Day 1 - Morning (3 hrs)

Understand: Concepts: AI, web, full-stack, database

Day 1 - Afternoon (3 hrs)

Build & Deploy: Tools, AI agents, live build

Day 2

Hackathon: Teams of 5

You understand how a web app works end to end

CampusBite is live on the internet

Live demo of your own AI-powered app

Day 1: Guided Foundation

Input-heavy, structured learning, step-by-step live build.

Day 2: Independent Creation

Output-heavy, team ideation, building custom solutions from scratch.

The day-to-day rhythm of development has fundamentally shifted.

Before AI

80% Writing Code
(Typing manually, copying from old projects)

20% Thinking
(Reviewing, designing, validating)

After AI

30% Writing Code
(Generated in seconds from plain-English descriptions)

70% Thinking
(Reviewing, designing, validating)

The Death of Boilerplate.

The value of a developer is shifting from typing lines of syntax to engineering quality, architecture, and security.

AI generates the code, but humans build the software.

What AI Does Well

- Generate and refactor code
- Explain complex logic
- Fix syntax bugs and write unit tests
- Write documentation

What AI Cannot Do

- Understand unclear business requirements
- Make architectural and system design decisions
- Understand company-specific rules
- Take responsibility for production systems

AI is a powerful assistant—not a replacement for understanding problems, making decisions, and validating solutions.

New Skills Every Developer Needs

Claude

Skill	★ Priority	Why it matters
1. Problem Solving	★★★★★	Break large problems into smaller tasks.
2. Logical Thinking	★★★★★	Understand algorithms and workflows.
3. Prompt Engineering	★★★★☆	Know how to communicate effectively with AI.
4. Code Review	★★★★★	AI generates code. You verify: Is it correct? Is it secure? Is it maintainable?
5. System Design	★★★★☆	Understand how software components work together.
6. Debugging	★★★★★	AI cannot always identify the real cause of complex production issues.
7. Testing	★★★★★	Validate that software works correctly and safely.
8. Git & Collaboration	★★★★☆	Modern software is built by teams, not individuals.
9. Cloud & DevOps	★★★★☆	Understand deployment, infrastructure, monitoring, and automation.
10. AI Tool Mastery	★★★★★	Work effectively with Claude Code, ChatGPT, GitHub Copilot, Gemini, Cursor, and MCP tools.

Top IT Companies Are Investing in Claude Code

ENTERPRISES CHOOSING ANTHROPIC CLAUDE

TCS TATA CONSULTANCY SERVICES
Infosys
accenture
Deloitte.
wipro
Cognizant
pwc
Capgemini
HCLTech
IBM
LTIMindtree
Tech Mahindra
ORACLE
ThoughtWorks
NTT DATA

Trusted by global enterprises to build, scale and innovate with Claude.

Understanding the foundational layers of the Agentic Era.

1. GenAI (The Creation)

Generative AI creates new content—text, code, or images—rather than just sorting existing data.

2. LLMs (The Brain)

Large Language Models are trained on massive amounts of text to predict and generate coherent code.

3. Tokens (The Currency)

LLMs break text into chunks called tokens, which dictate cost, speed, and how much the model can see at once.

4. Agents (The Action)

Software that plans multi-step tasks, uses tools, observes results, and adjusts course.

Where Claude Code fits into the AI coding landscape.

	Tool	Mechanism	Best Used For
1	ChatGPT / Gemini	Conversational AI in a browser.	Quick answers, isolated code snippets, learning concepts.
2	GitHub Copilot	Inline Autocomplete.	Speeding up typing for code you already know you want to write.
3	Cursor / Windsurf	AI-Native IDE.	An AI-first editing experience with fast in-editor edits.
4	Claude Code	Agentic / Multi-step.	Whole-project tasks, reading files, running commands, and fixing its own mistakes.

How we reach the finish line

Day 1 - Morning (3 hrs)

Understand

- GenAI, LLMs, Web mechanics, Full-stack, Database

Outcome: Understand how a web app works end to end.

Day 1 - Afternoon (3 hrs)

Build & Deploy

- VS Code, Git, Google Antigravity, Gemini agent, Vercel deploy

Outcome: Your app is live on the internet.

Day 2 - Full Day

Hackathon

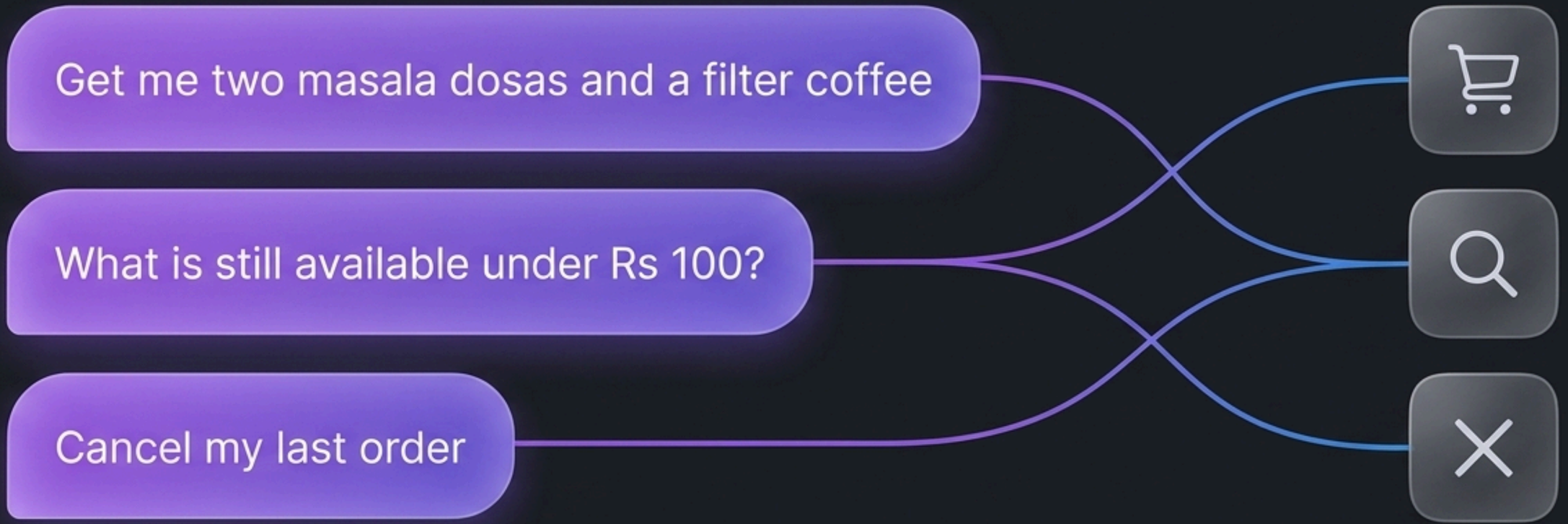
- Teams of 5, your own idea, same tech stack, live demo

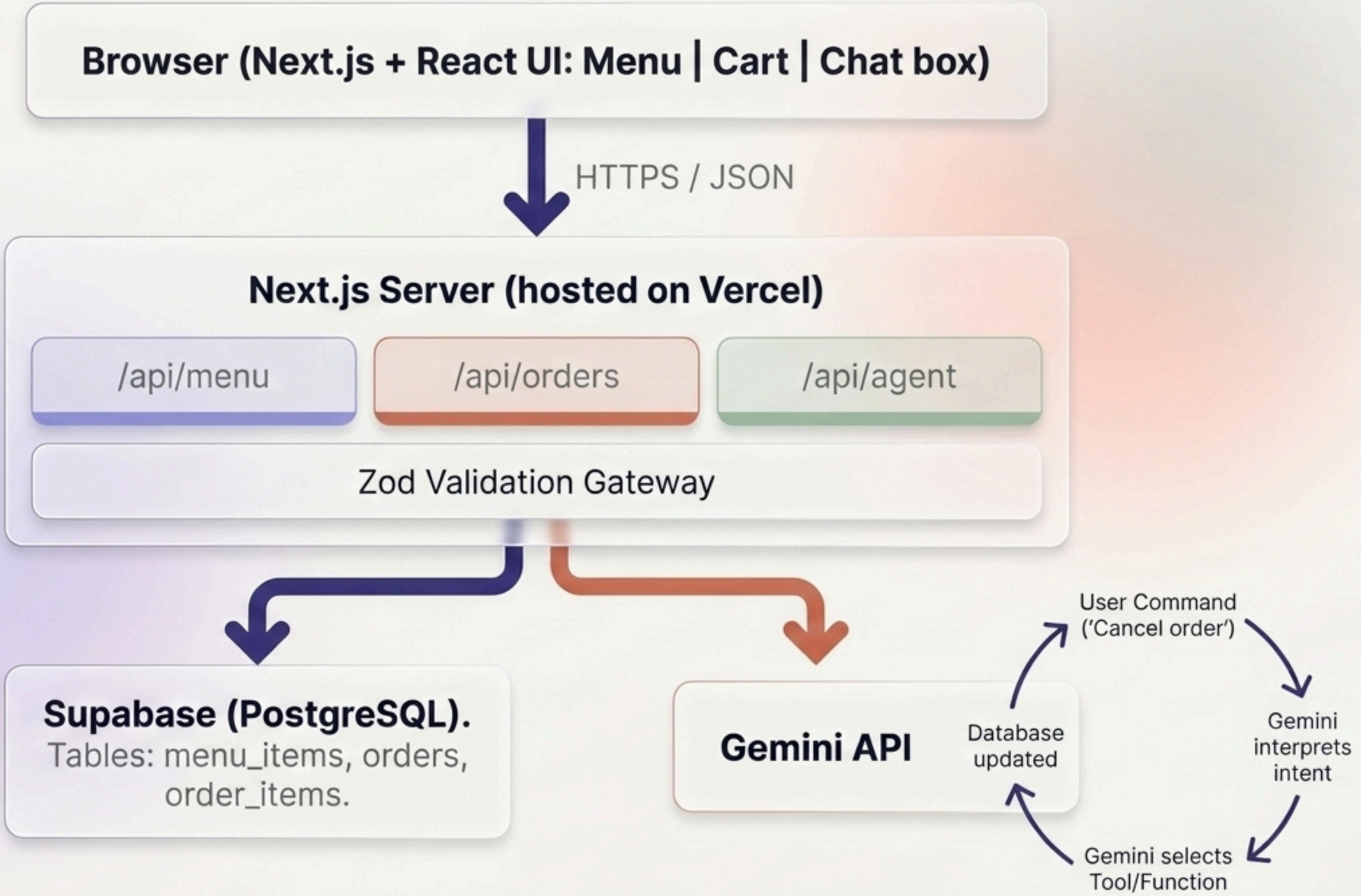
Outcome: Build and demo your own AI-powered app.

Our target build is CampusBite

A campus canteen ordering app designed to eliminate queues.
Students browse, order, and track; canteen staff manage and fulfill.

The magic layer is an AI assistant that acts as a smart waiter




Browser (Next.js + React UI: Menu | Cart | Chat box)
HTTPS / JSON
Next.js Server (hosted on Vercel)
/api/menu
/api/orders
/api/agent
Zod Validation Gateway
Supabase (PostgreSQL).
Tables: menu_items, orders, order_items.
Gemini API
User Command ('Cancel order')
Gemini interprets intent
Gemini selects Tool/Function
Database updated

The technology stack powering our application

Frontend

Next.js

React

TypeScript

Tailwind CSS

Backend

Next.js Route Handlers

Zod

Database

Supabase (PostgreSQL)

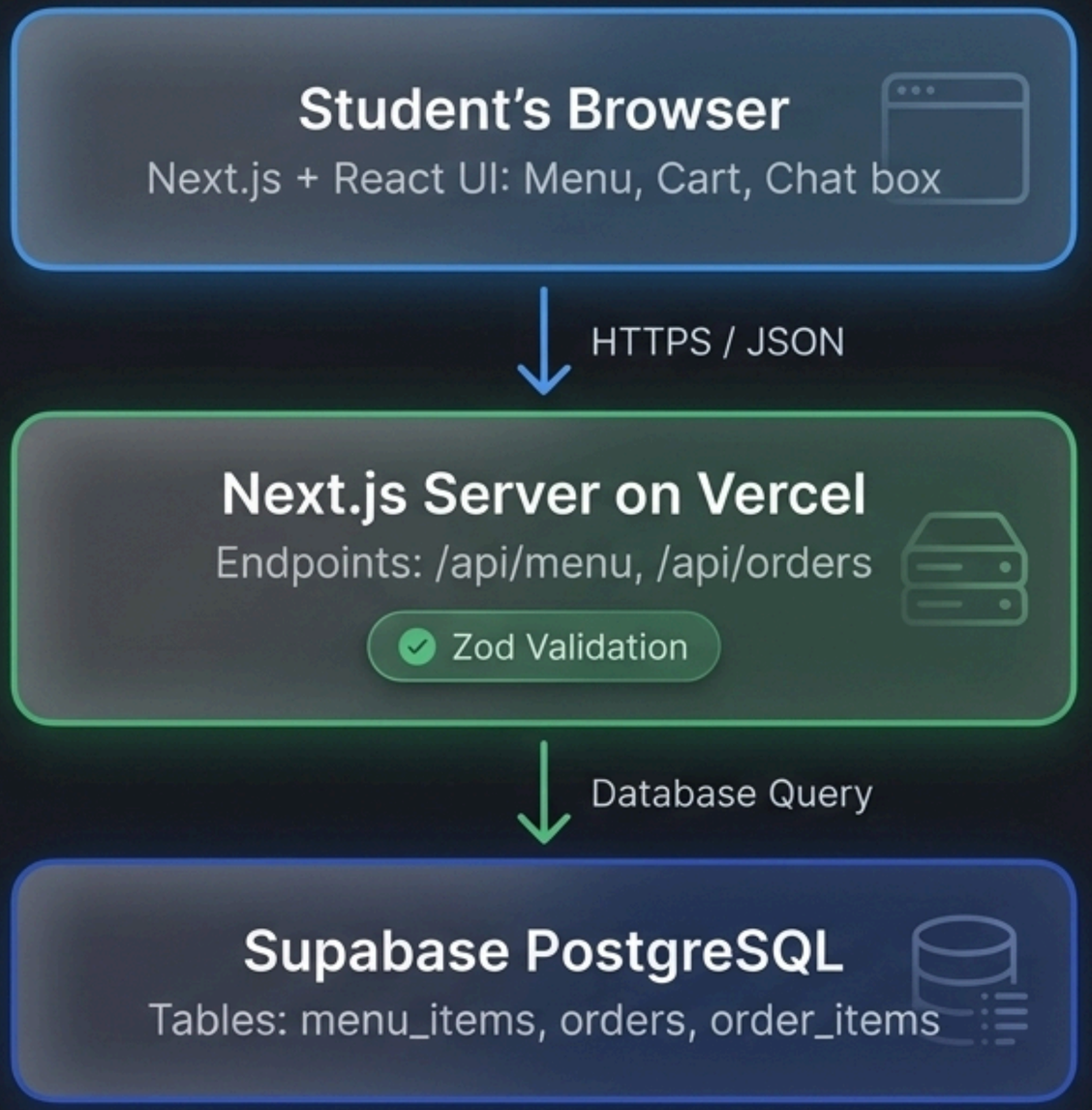
AI

Gemini API

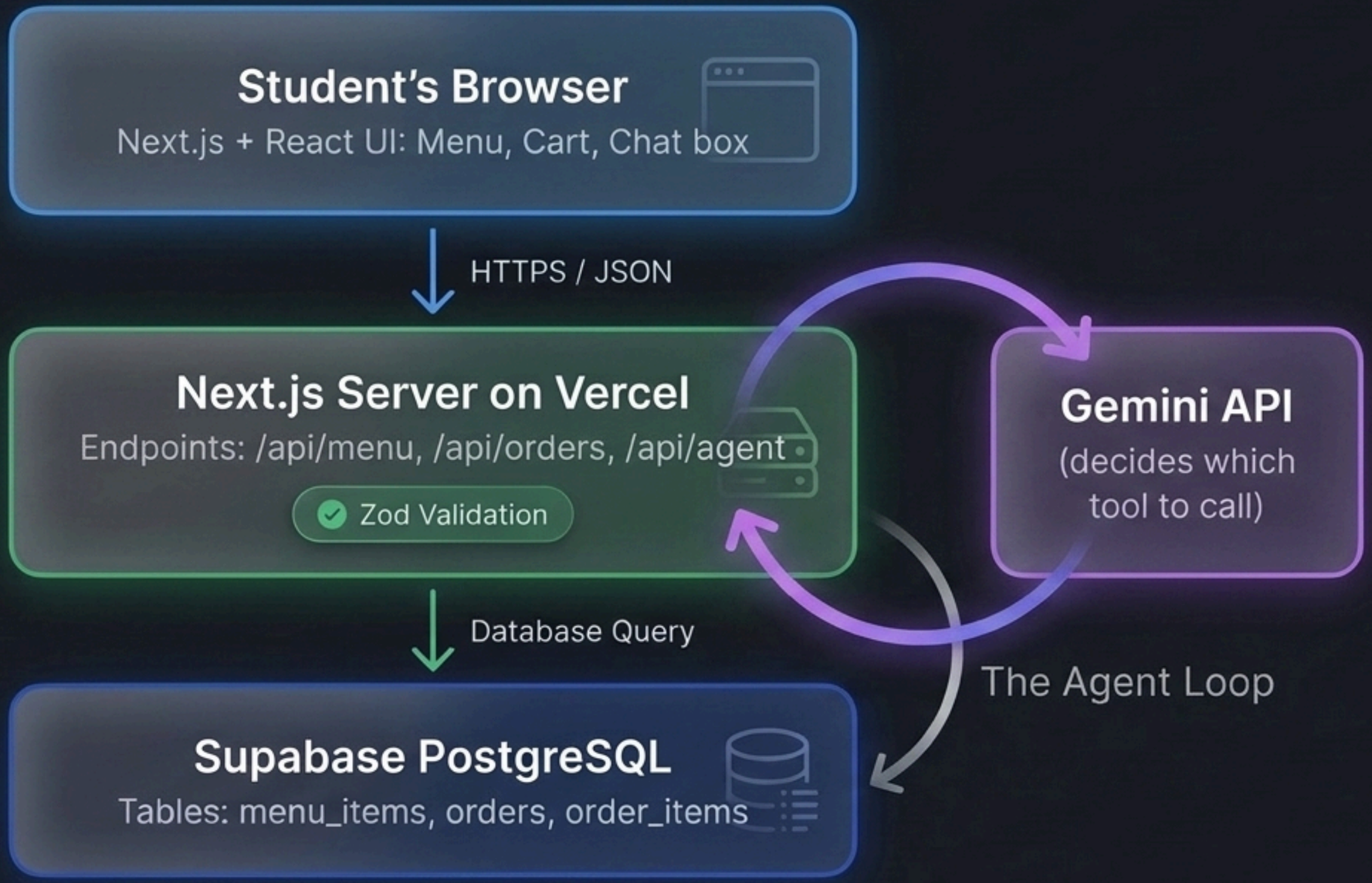
Google Antigravity

Hosting via Vercel

The standard flow of a modern web application



Integrating the Agent Loop into the server logic



A quick diagnostic of our starting point today

Raise your hand or vote on the poll link. If no hands go up for you, that is fine. Today is built for you.

I have written HTML/CSS/JavaScript before.

I have built a React or Next.js app.

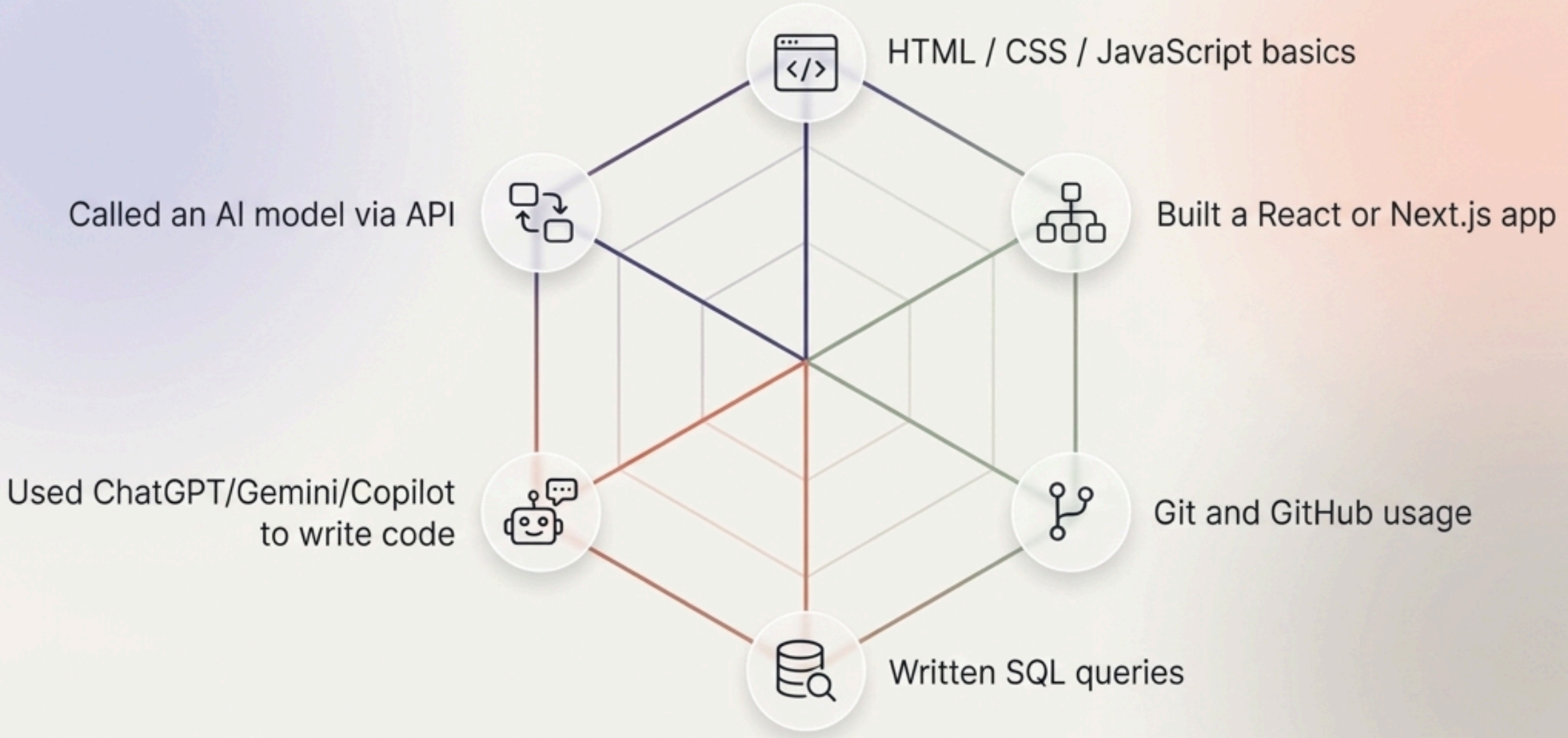
I have used Git and GitHub.

I have written SQL queries.

I have used ChatGPT, Gemini, or Copilot to write code.

I have called an AI model from my own code.

Calibrating our starting point: Which of these nodes have you unlocked?



Zero nodes unlocked? Today is built exactly for you. **Strong in a few?** Get ready to help your neighbors and lead your Hackathon team tomorrow.